

STATS 7022 - Data Science

Seminar 2: Linear Regression

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Bias and Variance of Estimators

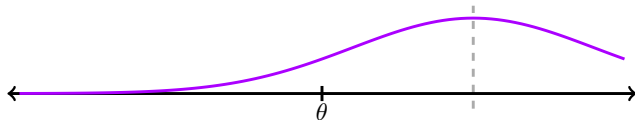
In courses like STATS 7107 Statistical Modelling and Inference, we discussed the bias and variance of estimators.

It can be helpful to visualise what this means.

Suppose we are considering several candidate estimators for a parameter θ .

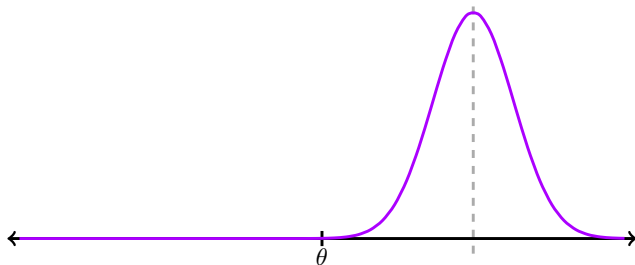
Bias and Variance of Estimators

An estimator may have large variance and large bias.



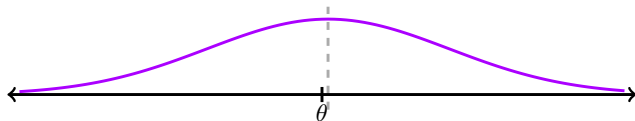
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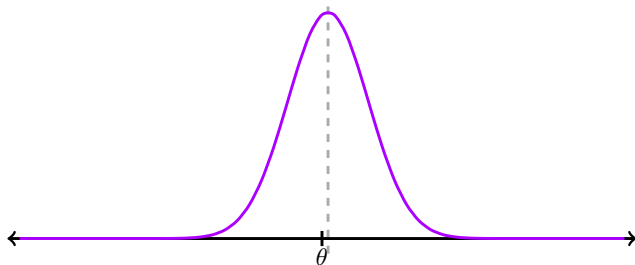
Bias and Variance of Estimators

An estimator may have large variance and small bias.



Bias and Variance of Estimators

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Simple Linear Regression

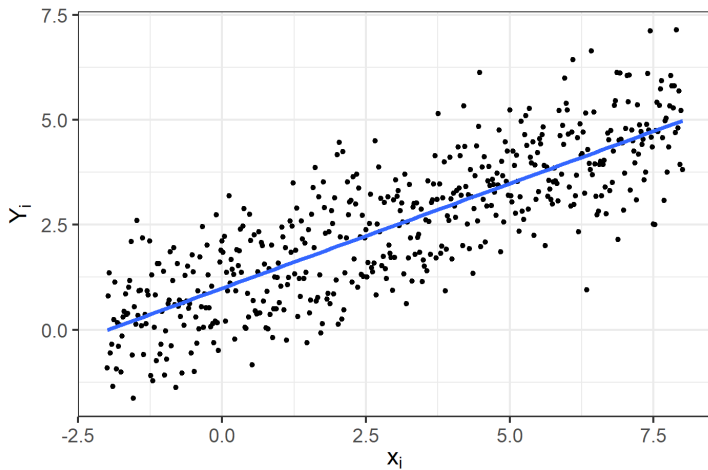
Consider data $(x_1, Y_1), \dots, (x_n, Y_n)$, with $x_i, Y_i \in \mathbb{R}$ for $i = 1, \dots, n$.

If a linear relationship exists between x_i and Y_i , we have the model

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

where $\epsilon_i \sim N(0, \sigma^2)$ independently for $i = 1, \dots, n$.

Simple Linear Regression



Simple Linear Regression

In practice, we may not know the *true* model.

Instead, we observe data $(x_1, y_1), \dots, (x_n, y_n)$ and use it to *estimate* the model parameters.

A simple linear regression model has two parameters:

- ▶ β_0 - the intercept
- ▶ β_1 - the slope

Simple Linear Regression

We *estimate* β_j as $\hat{\beta}_j$, where

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} \quad \text{and} \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}.$$

We define the quantities

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$$

and

$$S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y}).$$

Estimating the Slope ($\hat{\beta}_1$)

We would like to find the bias and MSE of $\hat{\beta}_1$.

First, we note that $\hat{\beta}_1$ is a *linear estimator* for β_1 .

We can write

$$\hat{\beta}_1 = \sum_{i=1}^n a_i Y_i$$

where

$$a_i = \frac{x_i - \bar{x}}{S_{xx}}.$$

Estimating the Slope ($\hat{\beta}_1$)

$$\begin{aligned} S_{xy} &= \sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y}) \\ &= \sum_{i=1}^n (x_i - \bar{x})Y_i - \sum_{i=1}^n (x_i - \bar{x})\bar{Y} \\ &= \sum_{i=1}^n (x_i - \bar{x})Y_i - \bar{Y} \sum_{i=1}^n (x_i - \bar{x}) \\ &= \sum_{i=1}^n (x_i - \bar{x})Y_i - \bar{Y} \left(\sum_{i=1}^n x_i - \sum_{i=1}^n \bar{x} \right) \\ &= \sum_{i=1}^n (x_i - \bar{x})Y_i - \bar{Y} \left(\sum_{i=1}^n x_i - n\bar{x} \right) \end{aligned}$$

Estimating the Slope ($\hat{\beta}_1$)

$$\begin{aligned} S_{xy} &= \sum_{i=1}^n (x_i - \bar{x}) Y_i - \bar{Y} \left(\sum_{i=1}^n x_i - n\bar{x} \right) \\ &= \sum_{i=1}^n (x_i - \bar{x}) Y_i - \bar{Y} \left(\sum_{i=1}^n x_i - n \frac{1}{n} \sum_{i=1}^n x_i \right) \\ &= \sum_{i=1}^n (x_i - \bar{x}) Y_i - \bar{Y} \left(\sum_{i=1}^n x_i - \sum_{i=1}^n x_i \right) \\ &= \sum_{i=1}^n (x_i - \bar{x}) Y_i \end{aligned}$$

Estimating the Slope ($\hat{\beta}_1$)

Thus,

$$\begin{aligned}\hat{\beta}_1 &= \frac{S_{xy}}{S_{xx}} \\&= \frac{1}{S_{xx}} \sum_{i=1}^n (x_i - \bar{x}) Y_i \\&= \sum_{i=1}^n \frac{(x_i - \bar{x})}{S_{xx}} Y_i \\&= \sum_{i=1}^n a_i Y_i, \text{ where } a_i = \frac{x_i - \bar{x}}{S_{xx}}.\end{aligned}$$

Estimating the Slope ($\hat{\beta}_1$)

What is the bias of $\hat{\beta}_1$?

$$\begin{aligned} E[\hat{\beta}_1] &= E \left[\sum_{i=1}^n a_i Y_i \right] \\ &= \sum_{i=1}^n a_i E[Y_i] \\ &= \sum_{i=1}^n a_i E[\beta_0 + \beta_1 x_i + \epsilon_i] \\ &= \sum_{i=1}^n a_i (\beta_0 + \beta_1 x_i + E[\epsilon_i]) \\ &= \sum_{i=1}^n a_i (\beta_0 + \beta_1 x_i) \end{aligned}$$

Estimating the Slope ($\hat{\beta}_1$)

$$\begin{aligned}E[\hat{\beta}_1] &= \sum_{i=1}^n a_i (\beta_0 + \beta_1 x_i) \\&= \sum_{i=1}^n a_i \beta_0 + \sum_{i=1}^n a_i \beta_1 x_i \\&= \beta_0 \sum_{i=1}^n a_i + \beta_1 \sum_{i=1}^n a_i x_i \\&= \beta_0 \sum_{i=1}^n \frac{(x_i - \bar{x})}{S_{xx}} + \beta_1 \sum_{i=1}^n \frac{(x_i - \bar{x})}{S_{xx}} x_i \\&= \frac{\beta_0}{S_{xx}} \sum_{i=1}^n (x_i - \bar{x}) + \frac{\beta_1}{S_{xx}} \sum_{i=1}^n (x_i - \bar{x}) x_i\end{aligned}$$

Estimating the Slope ($\hat{\beta}_1$)

By inspection, $\sum_{i=1}^n (x_i - \bar{x}) = 0$.

By similar arguments to those used for S_{xy} , it can be shown that

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})x_i.$$

Thus,

$$E[\hat{\beta}_1] = \frac{\beta_1}{S_{xx}} S_{xx} = \beta_1$$

so $\hat{\beta}_1$ is *unbiased* for β_1 .

Estimating the Slope ($\hat{\beta}_1$)

Recall that $MSE_{\hat{\beta}_1}(\beta_1) = \text{var}(\hat{\beta}_1) + b_{\hat{\beta}_1}(\beta_1)^2$.

Since $\hat{\beta}_1$ is unbiased for β_1 , we have $MSE_{\hat{\beta}_1}(\beta_1) = \text{var}(\hat{\beta}_1)$.

Thus, $MSE_{\hat{\beta}_1}(\beta_1) = \frac{\sigma^2}{S_{xx}}$.

Tasks

1. Show that $\text{var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}$.
2. Show that $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$ is a linear estimator. That is, show that

$$\hat{\beta}_0 = \sum_{i=1}^n b_i Y_i$$

for some constants b_i .

3. Find the bias and MSE of $\hat{\beta}_0$.

Task 1 Solution

$$\begin{aligned}\text{var}(\hat{\beta}_1) &= \text{var}\left(\sum_{i=1}^n a_i Y_i\right) \\&= \sum_{i=1}^n a_i^2 \text{var}(Y_i), \text{ by independence of } Y_i \\&= \sigma^2 \sum_{i=1}^n a_i^2 \\&= \sigma^2 \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{S_{xx}}\right)^2\end{aligned}$$

Task 1 Solution

$$\begin{aligned}\text{var}(\hat{\beta}_1) &= \sigma^2 \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{S_{xx}} \right)^2 \\ &= \frac{\sigma^2}{S_{xx}^2} \sum_{i=1}^n (x_i - \bar{x})^2 \\ &= \frac{\sigma^2 S_{xx}}{S_{xx}^2} \\ &= \frac{\sigma^2}{S_{xx}}.\end{aligned}$$

Task 2 Solution

$$\begin{aligned}\hat{\beta}_0 &= \bar{Y} - \hat{\beta}_1 \bar{x} \\&= \frac{1}{n} \sum_{i=1}^n Y_i - \sum_{i=1}^n a_i \bar{x} Y_i \\&= \sum_{i=1}^n \left(\frac{1}{n} - \frac{(x_i - \bar{x}) \bar{x}}{S_{xx}} \right) Y_i \\&= \sum_{i=1}^n b_i Y_i, \text{ where } b_i = \left(\frac{1}{n} - \frac{(x_i - \bar{x}) \bar{x}}{S_{xx}} \right).\end{aligned}$$

Task 3 Solution

Bias of $\hat{\beta}_0$:

$$\begin{aligned} E[\hat{\beta}_0] &= E \left[\sum_{i=1}^n b_i Y_i \right] \\ &= \sum_{i=1}^n b_i E[Y_i], \text{ by linearity of summation and expectation} \\ &= \sum_{i=1}^n b_i E[\beta_0 + \beta_1 x_i + \epsilon_i] \\ &= \sum_{i=1}^n b_i (\beta_0 + \beta_1 x_i), \text{ as } E[\epsilon_i] = 0 \\ &= \sum_{i=1}^n \left(\frac{1}{n} - \frac{(x_i - \bar{x})\bar{x}}{S_{xx}} \right) (\beta_0 + \beta_1 x_i) \end{aligned}$$

Task 3 Solution

$$\begin{aligned}E[\hat{\beta}_0] &= \sum_{i=1}^n \left(\frac{1}{n} - \frac{(x_i - \bar{x})\bar{x}}{S_{xx}} \right) (\beta_0 + \beta_1 x_i) \\&= \sum_{i=1}^n \frac{\beta_0}{n} - \frac{\beta_0 \bar{x}}{S_{xx}} \sum_{i=1}^n (x_i - \bar{x}) + \frac{\beta_1}{n} \sum_{i=1}^n x_i + \frac{\beta_1 \bar{x}}{S_{xx}} \sum_{i=1}^n (x_i - \bar{x}) x_i \\&= \beta_0 + 0 + \beta_1 \bar{x} - \beta_1 \bar{x} \\&= \beta_0\end{aligned}$$

Thus, $\hat{\beta}_0$ is unbiased for β_0 .

Task 3 Solution

MSE of $\hat{\beta}_0$:

$$\begin{aligned}MSE_{\hat{\beta}_0}(\beta_0) &= \text{var}(\hat{\beta}_0) + b_{\hat{\beta}_0}(\beta_0)^2 \\&= \text{var}(\hat{\beta}_0), \text{ as } \hat{\beta}_0 \text{ is unbiased for } \beta_0 \\&= \text{var}\left(\sum_{i=1}^n b_i Y_i\right) \\&= \sum_{i=1}^n b_i^2 \text{var}(Y_i), \text{ by independence of } Y_i \\&= \sigma^2 \sum_{i=1}^n b_i^2\end{aligned}$$

Task 3 Solution

$$\begin{aligned}MSE_{\hat{\beta}_0}(\beta_0) &= \sigma^2 \sum_{i=1}^n b_i^2 \\&= \sigma^2 \sum_{i=1}^n \left(\frac{1}{n} - \frac{(x_i - \bar{x})\bar{x}}{S_{xx}} \right)^2 \\&= \sigma^2 \left(\sum_{i=1}^n \frac{1}{n^2} - \frac{2\bar{x}}{nS_{xx}} \sum_{i=1}^n (x_i - \bar{x}) + \frac{\bar{x}^2}{S_{xx}^2} \sum_{i=1}^n (x_i - \bar{x})^2 \right) \\&= \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right).\end{aligned}$$

Best Linear Unbiased Estimation

If an estimator T for a parameter θ is the *best linear unbiased estimator* (BLUE) for θ , then:

- ▶ T has the lowest variance among all linear unbiased estimators for θ (*best*).
- ▶ T is a *linear* estimator for θ .
- ▶ T is an *unbiased* estimator for θ .

Best Linear Unbiased Estimation

By *best*, we mean:

“If T is the best linear unbiased estimator for a parameter θ , then for any other linear unbiased estimator $\tilde{T}$ for θ , $\text{var}(\tilde{T}) \geq \text{var}(T)$.”

BLUE for β_1

Consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

where $\epsilon_i \sim N(0, \sigma^2)$ independently for $i = 1, \dots, n$.

Then $\hat{\beta}_1$ is the *best linear unbiased estimator* (BLUE) for β_1 .

BLUE for β_1

We have previously shown that $\hat{\beta}_1$ is a *linear* estimator, written in the form $\hat{\beta}_1 = \sum_{i=1}^n a_i Y_i$.

We have also shown that $\hat{\beta}_1$ is *unbiased* for β_1 ; that is, $E[\hat{\beta}_1] = \beta_1$ and $b_{\hat{\beta}_1}(\beta_1) = 0$.

So, to show that $\hat{\beta}_1$ is the BLUE for β_1 , we need to show that $\hat{\beta}_1$ is *best*.

BLUE for β_1

Showing that $\hat{\beta}_1$ is best among all linear unbiased estimators for β_1 requires three steps:

1. Consider another linear unbiased estimator $\tilde{\beta}_1$ for β_1 . Derive some properties for $\tilde{\beta}_1$.
2. Add 0 (this is a surprisingly powerful and effective trick!)
3. Show that the $\text{cov}(\tilde{\beta}_1 - \hat{\beta}_1, \hat{\beta}_1) = 0$.

$\hat{\beta}_1$ is Best

Let $\tilde{\beta}_1$ be a linear unbiased estimator for β_1 .

Then $\tilde{\beta}_1 = \sum_{i=1}^n c_i Y_i$, and $E[\tilde{\beta}_1] = \beta_1$.

It follows that $\sum_{i=1}^n c_i = 0$ and $\sum_{i=1}^n c_i x_i = 1$.

$\hat{\beta}_1$ is Best

$$\begin{aligned}\beta_1 &= E[\tilde{\beta}_1] \\&= E\left[\sum_{i=1}^n c_i Y_i\right] \\&= \sum_{i=1}^n c_i E[Y_i], \text{ by linearity of expectation and summation} \\&= \sum_{i=1}^n c_i (\beta_0 + \beta_1 x_i + E[\epsilon_i]) \\&= \beta_0 \sum_{i=1}^n c_i + \beta_1 \sum_{i=1}^n c_i x_i, \text{ as } E[\epsilon_i] = 0 \\&\Rightarrow \sum_{i=1}^n c_i = 0 \text{ and } \sum_{i=1}^n c_i x_i = 1 \text{ (by equating coefficients)}\end{aligned}$$

$\hat{\beta}_1$ is Best

Consider the variance of $\tilde{\beta}_1$.

$$\begin{aligned}\text{var}(\tilde{\beta}_1) &= \text{var}(\tilde{\beta}_1 - \hat{\beta}_1 + \hat{\beta}_1) \\ &= \text{var}(\tilde{\beta}_1 - \hat{\beta}_1) + \text{var}(\hat{\beta}_1) + 2 \text{cov}(\tilde{\beta}_1 - \hat{\beta}_1, \hat{\beta}_1) \\ &\quad \text{by the definition of variance.}\end{aligned}$$

$\hat{\beta}_1$ is Best

Consider the “cross-term”:

$$\begin{aligned}\text{cov}(\tilde{\beta}_1 - \hat{\beta}_1, \hat{\beta}_1) &= \text{cov}\left(\sum_{i=1}^n c_i Y_i - \sum_{i=1}^n a_i Y_i, \sum_{j=1}^n a_j Y_j\right) \\&= \text{cov}\left(\sum_{i=1}^n (c_i - a_i) Y_i, \sum_{j=1}^n a_j Y_j\right) \\&= \sum_{i,j=1}^n (c_i - a_i) a_j \text{cov}(Y_i, Y_j)\end{aligned}$$

By independence, we have $\text{cov}(Y_i, Y_j) = \begin{cases} \text{var}(Y_i) & \text{if } i = j \\ 0 & \text{otherwise.} \end{cases}$

$\hat{\beta}_1$ is Best

$$\begin{aligned}\text{cov}(\tilde{\beta}_1 - \hat{\beta}_1, \hat{\beta}_1) &= \sum_{i,j=1}^n (c_i - a_i) a_j \text{cov}(Y_i, Y_j) \\ &= \sum_{i=1}^n (c_i - a_i) a_i \text{var}(Y_i) \\ &= \sum_{i=1}^n (c_i - a_i) a_i \sigma^2 \\ &= \sigma^2 \sum_{i=1}^n c_i a_i - \sigma^2 \sum_{i=1}^n a_i^2\end{aligned}$$

$\hat{\beta}_1$ is Best

Recall that if $\hat{\beta}_1 = \sum_{i=1}^n a_i Y_i$, then $a_i = \frac{x_i - \bar{x}}{S_{xx}}$.

$$\begin{aligned}\text{cov}(\tilde{\beta}_1 - \hat{\beta}_1, \hat{\beta}_1) &= \sigma^2 \sum_{i=1}^n c_i a_i - \sigma^2 \sum_{i=1}^n a_i^2 \\&= \sigma^2 \sum_{i=1}^n \frac{c_i (x_i - \bar{x})}{S_{xx}} - \sigma^2 \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{S_{xx}^2} \\&= \frac{\sigma^2}{S_{xx}} \sum_{i=1}^n c_i x_i - \frac{\sigma^2 \bar{x}}{S_{xx}} \sum_{i=1}^n c_i - \frac{\sigma^2}{S_{xx}^2} \sum_{i=1}^n (x_i - \bar{x})^2 \\&= \frac{\sigma^2}{S_{xx}} - 0 - \frac{\sigma^2}{S_{xx}^2} S_{xx} \\&= 0\end{aligned}$$

$\hat{\beta}_1$ is Best

Thus,

$$\begin{aligned}\text{var}(\tilde{\beta}_1) &= \text{var}(\tilde{\beta}_1 - \hat{\beta}_1) + \text{var}(\hat{\beta}_1) + 2 \text{cov}(\tilde{\beta}_1 - \hat{\beta}_1, \hat{\beta}_1) \\ &= \text{var}(\tilde{\beta}_1 - \hat{\beta}_1) + \text{var}(\hat{\beta}_1) \\ &\geq \text{var}(\hat{\beta}_1)\end{aligned}$$

Hence, $\hat{\beta}_1$ has the lowest variance among all linear unbiased estimators for β_1 .

Hence, $\hat{\beta}_1$ is the BLUE for β_1 .

Tasks

4. Hence, show that $\hat{\beta}_0$ is the BLUE for β_0 .

Hints:

- ▶ Consider some linear unbiased estimator $\tilde{\beta}_0 = \sum_{i=1}^n d_i Y_i$.
- ▶ Show that $\sum_{i=1}^n d_i = 1$ and $\sum_{i=1}^n d_i x_i = 0$.
- ▶ Show that $\text{cov}(\tilde{\beta}_0 - \hat{\beta}_0, \hat{\beta}_0) = 0$.

Task 4 Solution

Let $\tilde{\beta}_0$ be a linear unbiased estimator for β_0 .

Then $\tilde{\beta}_0 = \sum_{i=1}^n d_i Y_i$, and $E[\tilde{\beta}_0] = \beta_0$.

It follows that $\sum_{i=1}^n d_i = 1$ and $\sum_{i=1}^n d_i x_i = 0$.

Task 4 Solution

$$\begin{aligned}\beta_0 &= E[\tilde{\beta}_0] \\&= E\left[\sum_{i=1}^n d_i Y_i\right] \\&= \sum_{i=1}^n d_i E[Y_i], \text{ by linearity of expectation and summation} \\&= \sum_{i=1}^n d_i (\beta_0 + \beta_1 x_i + E[\epsilon_i]) \\&= \beta_0 \sum_{i=1}^n d_i + \beta_1 \sum_{i=1}^n d_i x_i, \text{ as } E[\epsilon_i] = 0 \\&\Rightarrow \sum_{i=1}^n d_i = 1 \text{ and } \sum_{i=1}^n d_i x_i = 0 \text{ (by equating coefficients)}\end{aligned}$$

Task 4 Solution

Consider the variance of $\tilde{\beta}_0$.

$$\begin{aligned}\text{var}(\tilde{\beta}_0) &= \text{var}(\tilde{\beta}_0 - \hat{\beta}_0 + \hat{\beta}_0) \\ &= \text{var}(\tilde{\beta}_0 - \hat{\beta}_0) + \text{var}(\hat{\beta}_0) + 2 \text{cov}(\tilde{\beta}_0 - \hat{\beta}_0, \hat{\beta}_0) \\ &\quad \text{by the definition of variance.}\end{aligned}$$

Task 4 Solution

Consider the “cross-term”:

$$\begin{aligned}\text{cov}(\tilde{\beta}_0 - \hat{\beta}_0, \hat{\beta}_0) &= \text{cov}\left(\sum_{i=1}^n d_i Y_i - \sum_{i=1}^n b_i Y_i, \sum_{j=1}^n b_j Y_j\right) \\&= \text{cov}\left(\sum_{i=1}^n (d_i - b_i) Y_i, \sum_{j=1}^n b_j Y_j\right) \\&= \sum_{i,j=1}^n (d_i - b_i) b_j \text{cov}(Y_i, Y_j)\end{aligned}$$

By independence, we have $\text{cov}(Y_i, Y_j) = \begin{cases} \text{var}(Y_i) & \text{if } i = j \\ 0 & \text{otherwise.} \end{cases}$

Task 4 Solution

$$\begin{aligned}\text{cov}(\tilde{\beta}_0 - \hat{\beta}_0, \hat{\beta}_0) &= \sum_{i,j=1}^n (d_i - b_i) b_j \text{cov}(Y_i, Y_j) \\&= \sum_{i=1}^n (d_i - b_i) b_i \text{var}(Y_i) \\&= \sum_{i=1}^n (d_i - b_i) b_i \sigma^2 \\&= \sigma^2 \sum_{i=1}^n d_i b_i - \sigma^2 \sum_{i=1}^n b_i^2\end{aligned}$$

Task 4 Solution

Recall from seminar 2 that if $\hat{\beta}_0 = \sum_{i=1}^n b_i Y_i$, then

$$b_i = \left(\frac{1}{n} - \frac{(x_i - \bar{x})\bar{x}}{S_{xx}} \right).$$

$$\begin{aligned} & \text{cov}(\tilde{\beta}_0 - \hat{\beta}_0, \hat{\beta}_0) \\ &= \sigma^2 \sum_{i=1}^n d_i b_i - \sigma^2 \sum_{i=1}^n b_i^2 \\ &= \sigma^2 \sum_{i=1}^n d_i \left(\frac{1}{n} - \frac{(x_i - \bar{x})\bar{x}}{S_{xx}} \right) - \sigma^2 \sum_{i=1}^n \left(\frac{1}{n} - \frac{(x_i - \bar{x})\bar{x}}{S_{xx}} \right)^2 \end{aligned}$$

Task 4 Solution

Consider

$$\begin{aligned}& \sum_{i=1}^n d_i \left(\frac{1}{n} - \frac{(x_i - \bar{x})\bar{x}}{S_{xx}} \right) \\&= \frac{1}{n} \sum_{i=1}^n d_i - \frac{\bar{x}}{S_{xx}} \sum_{i=1}^n d_i x_i + \frac{\bar{x}^2}{S_{xx}} \sum_{i=1}^n d_i \\&= \frac{1}{n} - 0 + \frac{\bar{x}^2}{S_{xx}} \\&= \frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\end{aligned}$$

Task 4 Solution

Consider

$$\sum_{i=1}^n \left(\frac{1}{n} - \frac{(x_i - \bar{x})\bar{x}}{S_{xx}} \right)^2$$

Task 4 Solution

Consider

$$\begin{aligned} & \sum_{i=1}^n \left(\frac{1}{n} - \frac{(x_i - \bar{x})\bar{x}}{S_{xx}} \right)^2 \\ &= \sum_{i=1}^n \frac{1}{n^2} - \end{aligned}$$

Task 4 Solution

Consider

$$\begin{aligned} & \sum_{i=1}^n \left(\frac{1}{n} - \frac{(x_i - \bar{x})\bar{x}}{S_{xx}} \right)^2 \\ &= \sum_{i=1}^n \frac{1}{n^2} - \frac{2\bar{x}}{nS_{xx}} \sum_{i=1}^n (x_i - \bar{x}) + \end{aligned}$$

Task 4 Solution

Consider

$$\begin{aligned} & \sum_{i=1}^n \left(\frac{1}{n} - \frac{(x_i - \bar{x})\bar{x}}{S_{xx}} \right)^2 \\ &= \sum_{i=1}^n \frac{1}{n^2} - \frac{2\bar{x}}{nS_{xx}} \sum_{i=1}^n (x_i - \bar{x}) + \frac{\bar{x}^2}{S_{xx}^2} \sum_{i=1}^n (x_i - \bar{x})^2 \end{aligned}$$

Task 4 Solution

Consider

$$\begin{aligned} & \sum_{i=1}^n \left(\frac{1}{n} - \frac{(x_i - \bar{x})\bar{x}}{S_{xx}} \right)^2 \\ &= \sum_{i=1}^n \frac{1}{n^2} - \frac{2\bar{x}}{nS_{xx}} \sum_{i=1}^n (x_i - \bar{x}) + \frac{\bar{x}^2}{S_{xx}^2} \sum_{i=1}^n (x_i - \bar{x})^2 \\ &= \frac{1}{n} - \end{aligned}$$

Task 4 Solution

Consider

$$\begin{aligned} & \sum_{i=1}^n \left(\frac{1}{n} - \frac{(x_i - \bar{x})\bar{x}}{S_{xx}} \right)^2 \\ &= \sum_{i=1}^n \frac{1}{n^2} - \frac{2\bar{x}}{nS_{xx}} \sum_{i=1}^n (x_i - \bar{x}) + \frac{\bar{x}^2}{S_{xx}^2} \sum_{i=1}^n (x_i - \bar{x})^2 \\ &= \frac{1}{n} - 0 + \end{aligned}$$

Task 4 Solution

Consider

$$\begin{aligned} & \sum_{i=1}^n \left(\frac{1}{n} - \frac{(x_i - \bar{x})\bar{x}}{S_{xx}} \right)^2 \\ &= \sum_{i=1}^n \frac{1}{n^2} - \frac{2\bar{x}}{nS_{xx}} \sum_{i=1}^n (x_i - \bar{x}) + \frac{\bar{x}^2}{S_{xx}^2} \sum_{i=1}^n (x_i - \bar{x})^2 \\ &= \frac{1}{n} - 0 + \frac{\bar{x}^2}{S_{xx}^2} S_{xx} \end{aligned}$$

Task 4 Solution

Consider

$$\begin{aligned}& \sum_{i=1}^n \left(\frac{1}{n} - \frac{(x_i - \bar{x})\bar{x}}{S_{xx}} \right)^2 \\&= \sum_{i=1}^n \frac{1}{n^2} - \frac{2\bar{x}}{nS_{xx}} \sum_{i=1}^n (x_i - \bar{x}) + \frac{\bar{x}^2}{S_{xx}^2} \sum_{i=1}^n (x_i - \bar{x})^2 \\&= \frac{1}{n} - 0 + \frac{\bar{x}^2}{S_{xx}^2} S_{xx} \\&= \frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\end{aligned}$$

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Thus,

$$\begin{aligned} & \text{cov}(\tilde{\beta}_0 - \hat{\beta}_0, \hat{\beta}_0) \\ &= \sigma^2 \sum_{i=1}^n d_i \left(\frac{1}{n} - \frac{(x_i - \bar{x})\bar{x}}{S_{xx}} \right) - \sigma^2 \sum_{i=1}^n \left(\frac{1}{n} - \frac{(x_i - \bar{x})\bar{x}}{S_{xx}} \right)^2 \\ &= \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right) - \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right) \\ &= 0 \end{aligned}$$

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Thus,

$$\begin{aligned}\text{var}(\tilde{\beta}_0) &= \text{var}(\tilde{\beta}_0 - \hat{\beta}_0) + \text{var}(\hat{\beta}_0) + 2 \text{cov}(\tilde{\beta}_0 - \hat{\beta}_0, \hat{\beta}_0) \\ &= \text{var}(\tilde{\beta}_0 - \hat{\beta}_0) + \text{var}(\hat{\beta}_0) \\ &\geq \text{var}(\hat{\beta}_0)\end{aligned}$$

Hence, $\hat{\beta}_0$ has the lowest variance among all linear unbiased estimators for β_0 .

Hence, $\hat{\beta}_0$ is the BLUE for β_0 .